178060: biotite monzogranite, Sandstone

Location and sampling

SANDSTONE (SG 50-16), SANDSTONE (2742)
MGA Zone 50, 731420E 6914200N

Sampled on 30 October 2002

The sample was taken from a fractured block, 0.5 m high and 20 cm thick, 200 m south of an east–west access track and 13.5 km north-northeast of Sandstone.

Tectonic unit/relations

The sample is a pale greenish-grey, fine- to medium- and even-grained, foliated biotite monzogranite in the Edale Shear Zone, Southern Cross Granite–Greenstone Terrane, Yilgarn Craton (Chen, 2005). The foliation trends to the northwest and plunges 30° to the southeast. At the sampling site, abundant thick white pegmatite dykes up to 2 cm thick are aligned parallel to the foliation. The sample was collected to determine the age of deformed rocks in the Edale Shear Zone.

Petrographic description

The principal minerals are microcline (40 vol.%), plagioclase (30–35 vol.%), quartz (25 vol.%), biotite (2–3 vol.%), opaque oxide (≤1 vol.%), and titanite (≤1 vol.%), with trace amounts of accessory zircon, and secondary muscovite, epidote, chlorite, and clay–sericite representing retrograde metamorphism and alteration. This is a quartz-poor biotite monzogranite gneiss with a strong quartz fabric. Much of the quartz occurs in narrow lenses and ribbons parallel to the foliation, with the majority of the quartz c-axes at a low angle to the foliation. Most of the plagioclase and microcline grains are less than 1 mm in grain size and are anhedral, but there are some microcline grains to 2 mm in length and elongate parallel to the foliation. Biotite flakes up to 0.5 mm long define a strong schistosity. Granular opaque oxide is mostly less than 0.3 mm in grain size, and rare titanite reaches 1 mm. Altered grains, which could have been allanite, are scattered but lack epidote rims. Zircon crystals range from 50 to 200 µm in length and as much as 50 µm in width, with zoning visible in the largest crystals. Muscovite, carbonate, and minor epidote have formed within plagioclase grains as well as later fine-grained clay–sericite with or without clinozoisite alteration. The plagioclase could also be albitized. About 40 vol.% of the biotite is altered to chlorite and clay, rarely with fine-grained epidote or muscovite, but the remaining biotite is fresh. Minor carbonate is included in titanite. This is a foliated or gneissic monzogranite with probably low amphibolite-facies metamorphism and strong deformation, accompanied by recrystallization, as indicated by the quartz fabric.

Zircon morphology

The zircons isolated from this sample are typically yellowish-brown, dark brown or black, euhedral and elongate in shape, and between 50 × 80 µm and 160 × 80 µm in size. A majority of grains have faint internal zonation, but a minority are internally structureless. Many grains are metamict. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 23 October and 29 November 2003. The counter deadtime during both analysis sessions was 24 ns. During the first analysis session, four analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 0.912% (1σ). Analyses 1.1 to 4.1 were obtained during the first analysis session. Six analyses of the CZ3 standard were obtained during the second analysis session. Following deletion of one of these analyses as an outlier, the remaining five analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 0.733% (1σ). A calibration uncertainty of 1.0% (1σ) was applied to analyses of unknowns obtained during both analysis sessions. Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 5.1, 7.1, 8.1, and 2.2, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

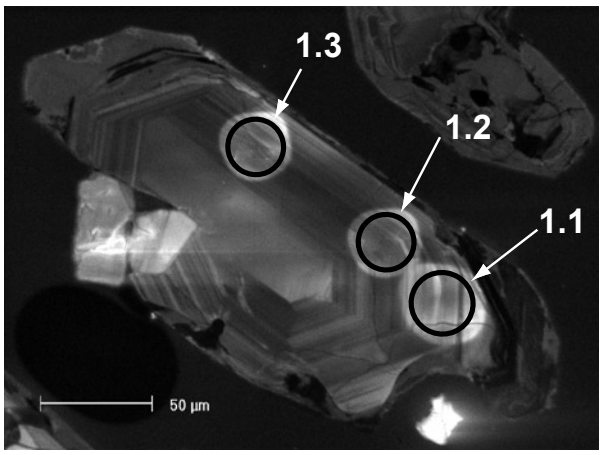
Results

Eighteen analyses were obtained from 13 zircons. Results are given in Table 1 and shown on a concordia plot in Figure 2.

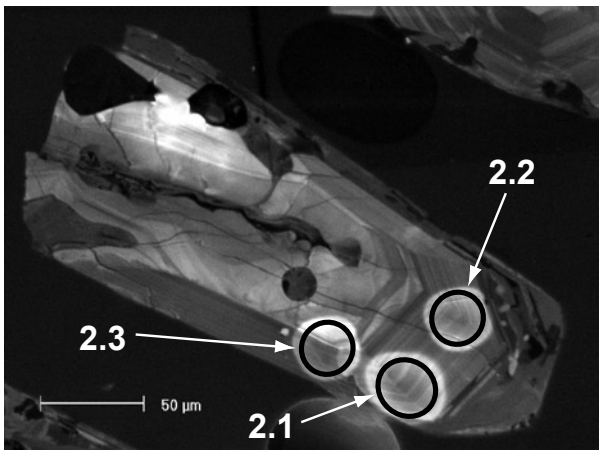
Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with both ancient and recent episodes of radiogenic-Pb loss. Ten concordant to highly discordant analyses of five zircons (1.1, 1.2, 1.3, 2.1, 2.2, 2.3, 3.1, 3.2, 5.1, 13.1) have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2683 ± 8 Ma (chi-squared = 0.44). The remaining analyses are highly discordant and indicate significantly younger $^{207}Pb/^{206}Pb$ dates.

The date of 2683 ± 8 Ma indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of 10 concordant to highly discordant analyses of five zircons is interpreted as the age of igneous crystallization of the monzogranite. The remaining analyses are interpreted to be of sites that have lost some radiogenic Pb.



(a)



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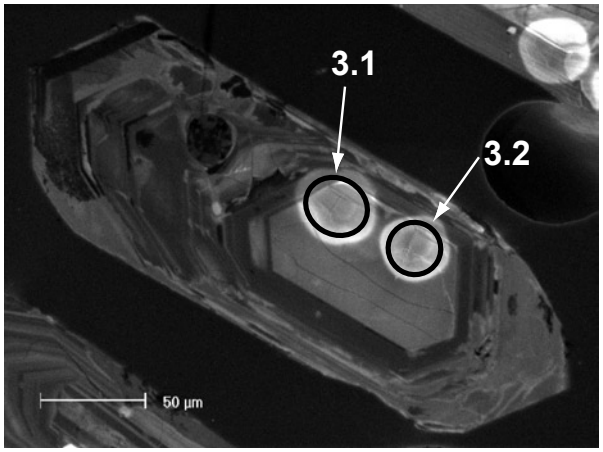

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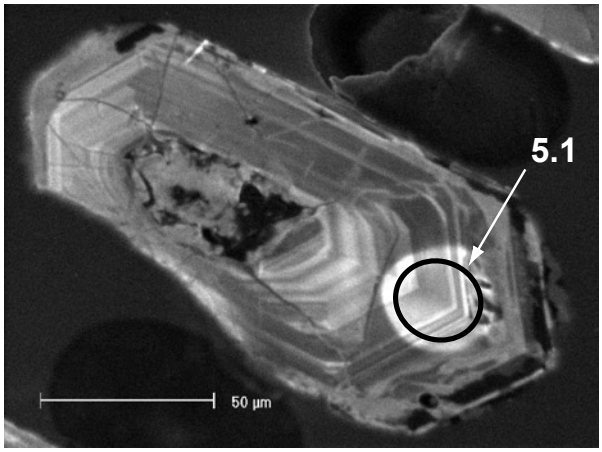

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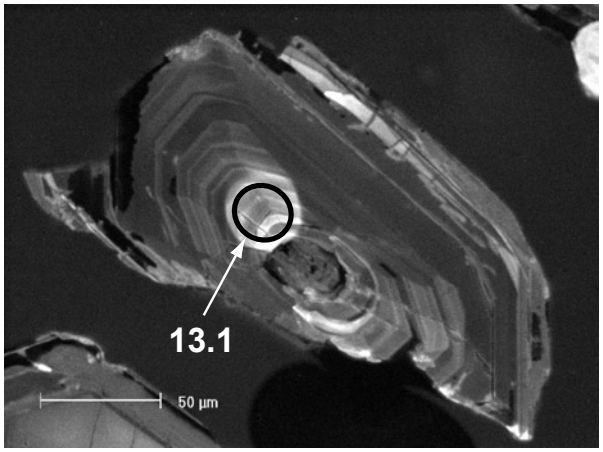

(e)

Figure 1. Cathodoluminescence images of representative zircons from sample 178060: biotite monzogranite, Sandstone

Table 1. Ion microprobe analytical results for zircons from sample 178060: biotite monzogranite, Sandstone

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	152	123	100	1.509	0.18508	0.00172	0.23852	0.00356	0.5134	0.0056	13.102	0.199	99	2 699	15
2.1	119	92	76	0.827	0.18355	0.00158	0.20606	0.00303	0.5201	0.0058	13.163	0.196	101	2 685	14
3.1	146	110	91	1.257	0.18451	0.00164	0.21221	0.00328	0.4992	0.0055	12.700	0.190	97	2 694	15
4.1	493	584	297	0.891	0.17596	0.00084	0.35110	0.00182	0.4461	0.0046	10.824	0.128	91	2 615	8
5.1	246	185	161	2.316	0.18244	0.00156	0.24358	0.00327	0.4956	0.0053	12.468	0.180	97	2 675	14
6.1	515	406	264	0.247	0.17155	0.00064	0.25924	0.00120	0.4138	0.0042	9.788	0.111	87	2 573	6
7.1	513	95	190	1.033	0.15989	0.00091	0.06907	0.00167	0.3369	0.0034	7.428	0.091	76	2 454	10
8.1	464	276	246	1.682	0.17481	0.00106	0.19450	0.00215	0.4247	0.0044	10.237	0.129	88	2 604	10
9.1	700	633	302	0.467	0.15947	0.00064	0.26385	0.00129	0.3471	0.0035	7.632	0.087	78	2 450	7
10.1	587	627	204	0.705	0.14272	0.00079	0.26719	0.00168	0.2806	0.0029	5.522	0.067	71	2 261	10
11.1	891	1 272	383	0.204	0.15943	0.00050	0.37789	0.00111	0.3229	0.0033	7.097	0.078	74	2 450	5
12.1	715	461	139	0.541	0.10800	0.00078	0.20407	0.00170	0.1695	0.0017	2.524	0.033	57	1 766	13
1.2	294	340	201	0.501	0.18233	0.00088	0.32347	0.00180	0.5193	0.0054	13.054	0.157	101	2 674	8
2.2	119	98	84	3.209	0.18427	0.00246	0.25884	0.00523	0.5136	0.0058	13.048	0.241	99	2 692	22
3.2	140	105	88	1.950	0.18378	0.00177	0.20815	0.00365	0.4938	0.0054	12.513	0.193	96	2 687	16
13.1	237	268	159	1.478	0.18425	0.00126	0.31031	0.00270	0.5007	0.0053	12.720	0.169	97	2 691	11
1.3	248	298	170	0.474	0.18274	0.00088	0.33342	0.00180	0.5193	0.0054	13.084	0.157	101	2 678	8
2.3	167	189	116	0.535	0.18355	0.00115	0.31250	0.00233	0.5334	0.0057	13.500	0.177	103	2 685	10

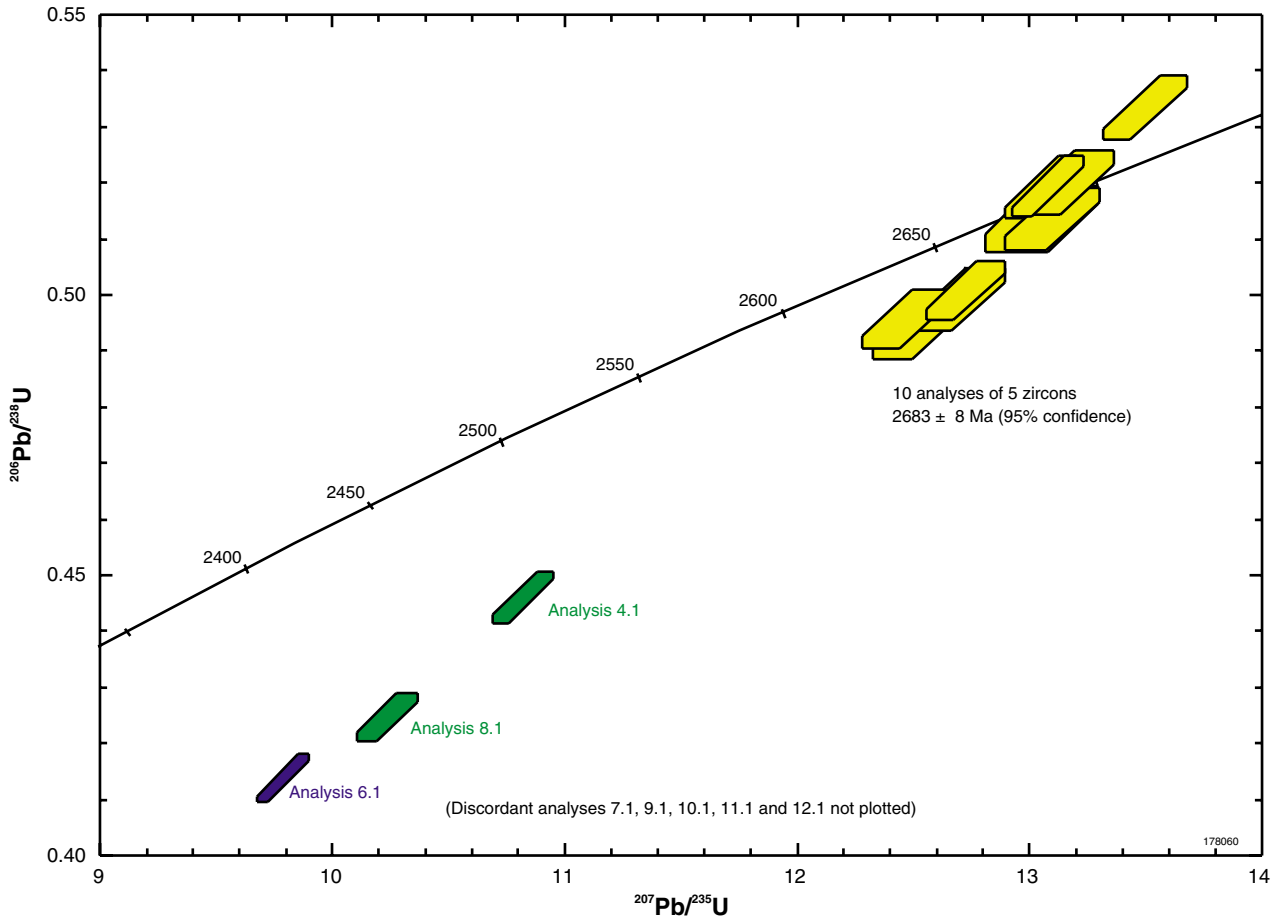

Figure 2. Concordia plot for sample 178060: biotite monzogranite, Sandstone

Recommended reference for this publication:

NELSON, D. R., 2005, 178060: biotite monzogranite, Sandstone; Geochronology dataset 587; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 29/11/2003; Data released: 30/06/2005